Warm-up / Do flashcards help?

Which evidence deserves more weight?

Name: _____ Date: _____

Suggested lesson time: 30 minutes. Work individually, then compare when invited.

Useful words

Testimonial: A person's report of their own experience; it may be sincere without showing what caused the experience.

Comparison group: A group used to judge what happens under a different condition, such as usual support instead of tutoring.

Random assignment: Using chance to allocate participants to groups. It reduces systematic group differences on average; it does not guarantee perfectly identical groups.

Outcome measure: The recorded result used to evaluate a claim, such as passing the same algebra exam.

Read the scenario

Invented scenario: A tutor claims flashcards improve vocabulary test scores. Report A quotes one learner: 'It really helped me.' Report B compares two groups of 20 learners with similar starting scores, but no random assignment: flashcards versus usual review sheets. Both groups review for the same time and take the same test. All scores are reported; the flashcard group has the higher average.

Produce: Write three sentences: choose a report, explain one strength in how its evidence was gathered, and name one remaining uncertainty.

Your steps

1. Choose which report gives stronger evidence for the tutor's claim.
2. Explain how either its comparison group or common test helps assess the claim.
3. State one reason the description does not establish the method's effect with certainty.

Before you finish: Can you explain your choice without treating a sincere testimonial as proof or treating a comparison as a guarantee?

Your response

Main activity / Two reports about algebra tutoring

Name: _____

Read the scenario

Invented scenario: A tutoring program claims it raises algebra pass rates. Report A quotes one volunteer saying tutoring helped. Report B follows 200 students randomly assigned to receive tutoring or usual support, uses the same exam, and reports all results: 70 of 100 tutored students pass, versus 55 of 100 others. Both reports are accurately described; use only these facts.

Produce: Write a choice, three reasons about the method, and one question about applying the result to another school.

Your steps

1. Choose the stronger evidence for the program's claim.
2. Give three reasons using the number of students, the comparison groups, and the outcome measure.
3. State one limit: what would you still want to know before assuming the result applies to another school?

Before you finish: Explain the evidence's method rather than treating a confident testimonial or a large number alone as decisive.

Your response

Exit ticket / Testing a spelling program

Name: _____

Read the scenario

Invented scenario: A school claims its new spelling program improves spelling test scores. Report A quotes one parent saying the program worked for their child. Report B records 80 pupils randomly assigned to the new program or usual spelling lessons for six weeks. Both groups receive equal lesson time and take the same final test; all results are reported. The new-program group's average is 16 out of 20, compared with 13 out of 20 for the usual-lessons group. No information about other schools is supplied.

Produce: Write the stronger report, three reasons tied to its method, and one question to ask before using the result in another school.

Your steps

1. Choose the stronger report for the stated claim.
2. Explain how the participants, comparison groups, and common outcome measure affect the strength of the evidence.
3. State a relevant detail you would check before assuming another school's pupils would benefit similarly.

Before you finish: Have you explained the method, kept your conclusion about the reported comparison, and avoided inventing a result for another school?

Your response